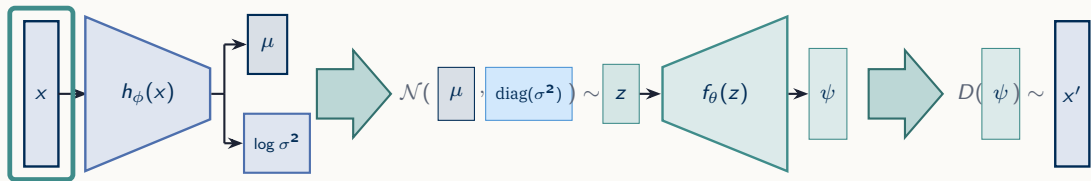
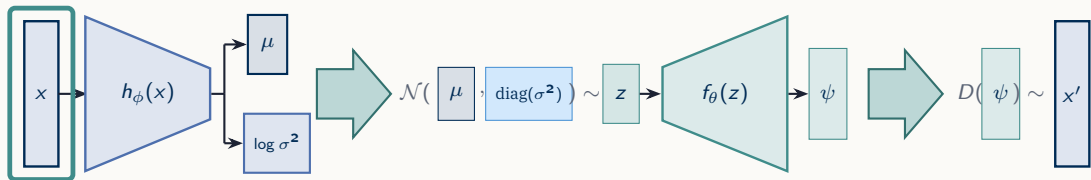


Input

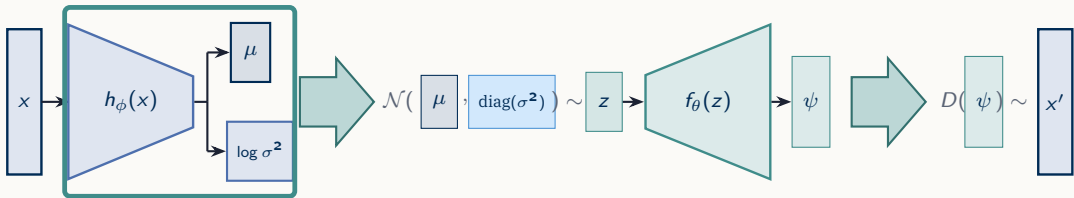


Input

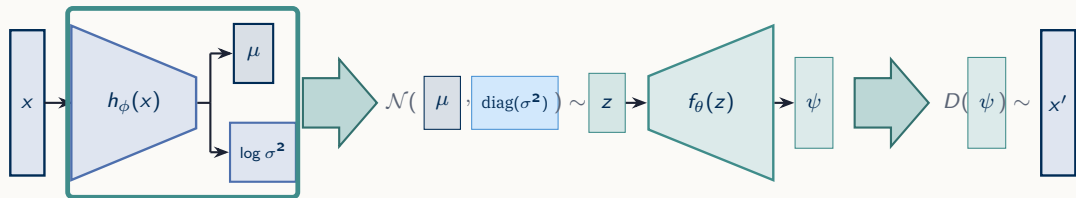


The **Variational Autoencoder** begins with an input sample x , which can be an **image**, **text**, or **any data**.

Encoder

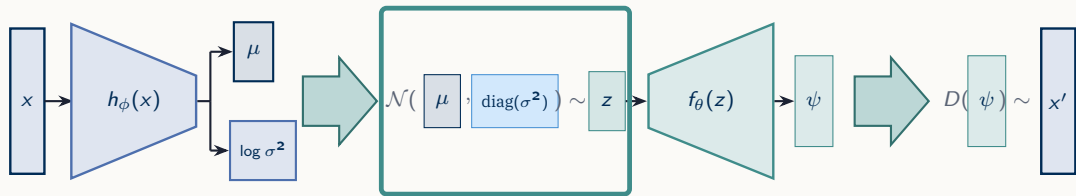


Encoder



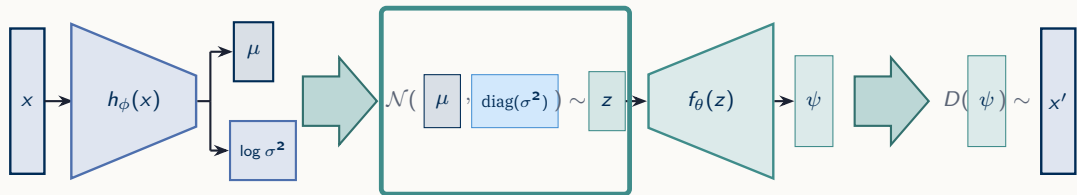
- The **Encoder** maps the input x into a **Latent Space Representation**.
- Outputs:
 - **Mean** (μ)
 - **Log Variance** ($\log \sigma^2$)

Latent Variable



z represents a compressed **probabilistic latent space**.

Latent Variable

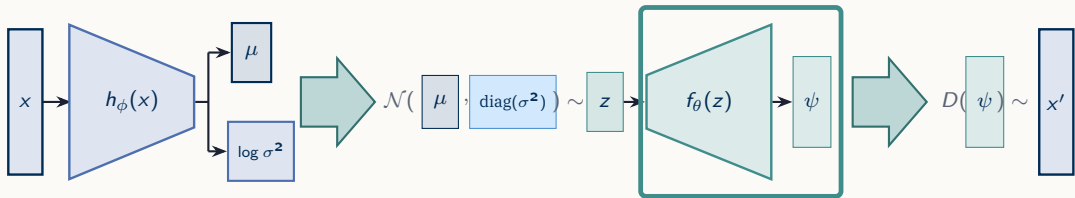


z represents a compressed **probabilistic latent space**.

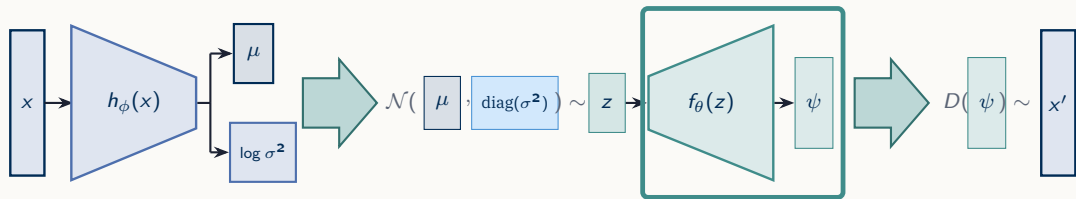
Instead of fixed values, z is drawn from a distribution defined by the encoder:

$$p(z | x) \sim \mathcal{N}(\mu, \sigma^2)$$

Decoder

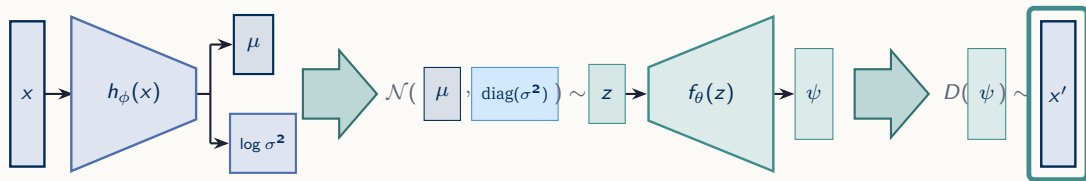


Decoder

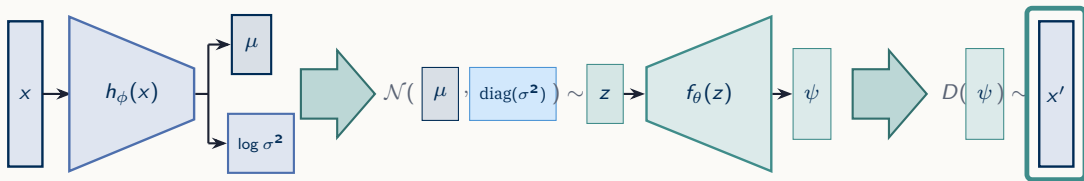


The decoder maps z to reconstructed data x' via $p_\theta(x | z)$.

Output: x'



Output: x'



The **VAE** generates data x' with the same input dimension as x using the latent variable z , i.e., $x' \in \mathbb{R}^d$ where d is the input dimension.

Reconstruction Loss

Reconstruction Loss: Measures the difference between the input x and the reconstructed output $\hat{x}$.

Reconstruction Loss

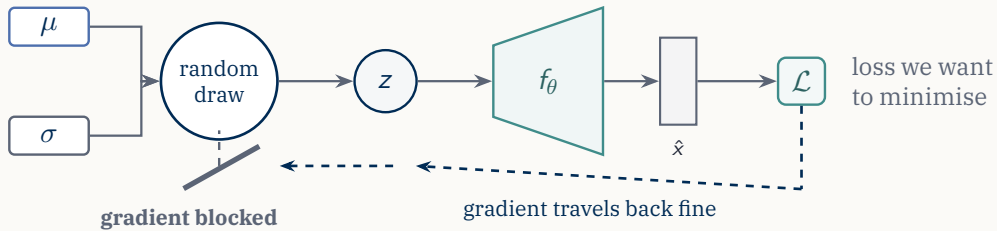
Reconstruction Loss: Measures the difference between the input x and the reconstructed output $\hat{x}$.

Mean Squared Error (MSE): measures the squared difference between x and $\hat{x}$.

$$\mathcal{L}_{\text{recon}} = \|x - \hat{x}\|^2$$

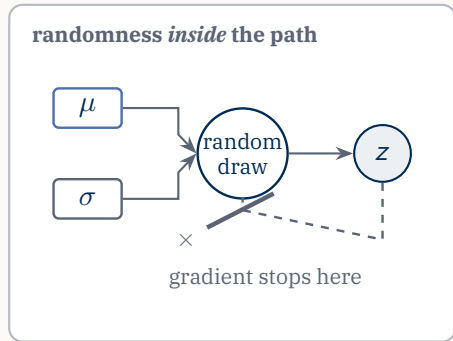
Binary Cross-Entropy (BCE): commonly used when the data is binary or normalized like pixel probabilities.

Why Sampling Breaks Training



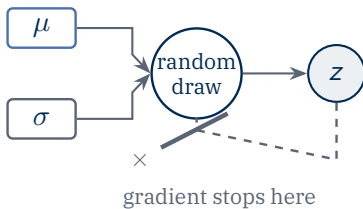
The loss gradient travels backward through the decoder, but **direct random sampling blocks it from reaching μ and σ .**

The Reparameterisation Trick

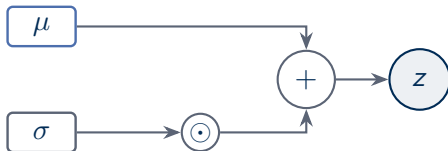


The Reparameterisation Trick

randomness *inside* the path

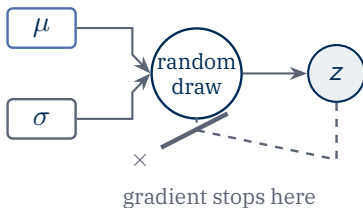


randomness moved *outside*

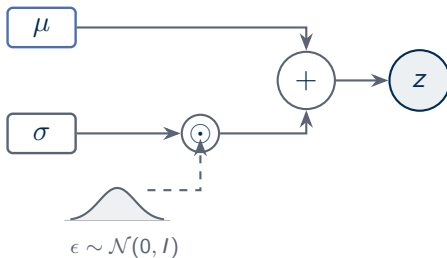


The Reparameterisation Trick

randomness *inside* the path

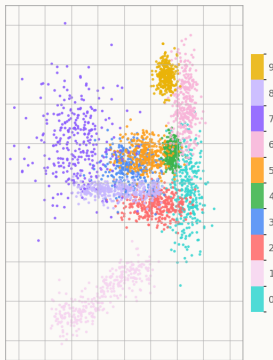


randomness moved *outside*



Reparameterise: $z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$

Only Reconstruction Loss



Problem: With only **reconstruction loss**, the model may reconstruct data well, but the latent space can become **irregular, discontinuous, and hard to sample from**. Random points may fall into weak regions, so generation becomes unreliable.

Kullback-Leibler (KL) Divergence

Measures the divergence between $q_\phi(z|x)$ and the prior $p(z)$, typically $\mathcal{N}(0, I)$.

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Regularization Purpose

Promotes a smooth, continuous latent space, fostering:

- **Continuity**
- **Completeness**

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KL Divergence Formula

$$\mathcal{L}_{KL} = D_{KL}(q_\phi(z|x) \parallel p(z)) = \int q_\phi(z|x) \log \frac{q_\phi(z|x)}{p(z)} dz$$

Evidence Lower Bound (ELBO)

ELBO: Overview

ELBO balances two objectives: **Reconstruction Quality** and **Latent Space Regularization**.

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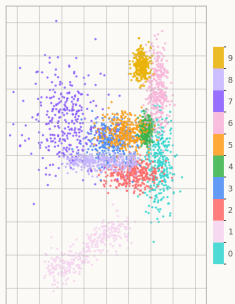
Goal

Maximizing ELBO is equivalent to minimizing the VAE loss:

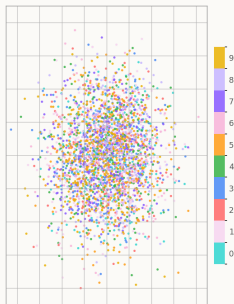
$$\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{KL}}$$

ELBO : Impact Visualization

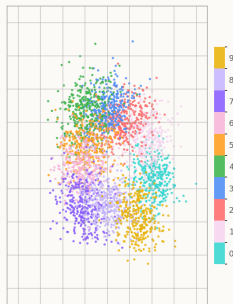
Only reconstruction loss



Only KL divergence



Reconstruction loss and KL divergence



ELBO combines both goals: $\text{ELBO} = \mathbb{E}_{q_{\phi}(z|x)}[\log p_{\theta}(x | z)] - D_{KL}(q_{\phi}(z | x) \| p(z))$

In practice: **reconstruct well** and **keep the latent space regular**.